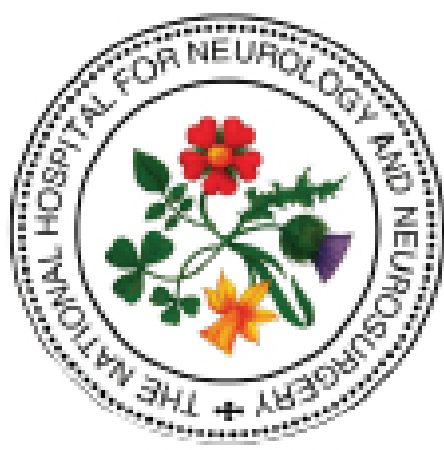




VASOPRESSIN

INDICATIONS FOR USE

Vasopressin (also called Argipressin) is indicated in the management of hypotension in noradrenaline resistant vasodilatory or septic shock and is used when noradrenaline is not achieving the required BP target, or as a noradrenaline sparing agent when the noradrenaline dose is high. It should be considered when noradrenaline requirement is $\geq 0.9\text{mcg/kg/min}$.

It should be administered as a continuous intravenous infusion via a central venous access device. If access is limited, vasopressin is compatible with noradrenaline. The half-life of vasopressin is around 10-20 minutes so **DOUBLE PUMPING IS NOT REQUIRED** when renewing the syringe

STABILITY and COMPATIBILITY

It should be used immediately after preparation but is stable in the syringe for 24 hours. Any unused drug should be discarded after 24 hours.

It is compatible with:

- Adrenaline
- Amiodarone
- Dobutamine
- Dopamine
- Noradrenaline

CAUTIONS and CONTRAINDICATIONS

- Cautions in epilepsy, migraine, asthma, heart failure and any state in which rapid addition to extracellular water may be hazardous
- Contra-indications include hypersensitivity to drug or components, chronic nephritis with nitrogen retention

ADVERSE EFFECTS

- Peripheral ischaemia
- Local or systemic allergic reactions
- Myocardial ischaemia, infarction and cardiac arrest, arrhythmias, bradycardia
- Gastrointestinal ischaemia, nausea, vomiting and abdominal cramps
- Fluid retention
- Rebound hypotension on discontinuation of the infusion – *wean slowly*

VASOPRESSIN USE in the SUPPORT of ORGAN DONORS

Following brain death, profound reduction in sympathetic outflow is associated with reduced cardiac output, vascular tone, haemodynamic instability. In addition to reducing catecholamine needs and reversing diabetes insipidus, vasopressin has been shown to limit pulmonary injury and decrease systemic inflammatory response.

INTRAVENOUS VASOPRESSIN for BP REFRACTORY to NORADRENALINE

PRESENTATION

Each 1mL vial contains 20 units of vasopressin
(Argipressin (20 units in 1mL) is synthetic vasopressin)

STEP 1

ADMINISTRATION

Dilute **20 units** (one vial) to **50mL** with sodium chloride 0.9% or dextrose 5% to a final concentration of **0.4 units/mL**

STEP 2

INFUSION

- **Range: 0.01 to 0.04 units/min** according to BP response
- **Start at 0.01 units/min (1.5mL/h)**, increasing at 0.01 units/min (1.5mL) increments half hourly, to a **maximum of 0.04 units/min (6mL/h)**
- Noradrenaline should be weaned slowly and titrated against target BP
- If vasopressin does not produce a response in BP and allow a reduction in noradrenaline after **2 hours** at maximum dose (6mL/h) discontinue vasopressin
- Vasopressin does **NOT** need to be 'double pumped'
- Doses of vasopressin >0.04 units/min are associated with increased adverse events without increased benefit

INFUSION RATE (mL/h) of 0.4 units/mL SOLUTION		
Units / min	Units / h	mL / hr
0.01	0.6	1.5
0.02	1.2	3
0.03	1.8	4.5
0.04	2.4	6

STEP 3

WEANING

- When vasopressin no longer required it should be slowly weaned down and off before the noradrenaline infusion is stopped
- Reduce the vasopressin infusion by half every 30 minutes down to 0.01 units/hour (1.5mL/hour) then stop

UKCPA Critical Care Society Intensive Care Society Standard Medication Concentrations for Continuous Infusions in Adult Critical Care			
The Intensive Care Society supports the adoption of standard concentrations and endorses the recommendations of a multi-professional group who have published a list of concentrations with wide acceptance in critical care for this purpose (1,2,3). Standardising concentrations represents a significant step towards improving both patient safety and efficient use of resources within critical care. It also facilitates the production of a national injectables guide to provide the end user with the information necessary to safely administer such medications (4). The adoption of these is recommended, not mandated. It is anticipated that pharmaceutical manufacturers will begin to prepare ready to use and ready to administer products based on this list.			
Medication	Concentration	Example Infusion Concentration	Central or Peripheral C / P
Morphine	1mg/mL	50mg in 50mL	C / P
Fentanyl	50micrograms/mL	100mg in 50mL	C / P
Alfentanil	500micrograms/mL	25mg in 50mL	C / P
Remifentanyl	50micrograms/mL	2mg in 40mL	C / P
Midazolam	100micrograms/mL	5mg in 50mL	C / P
Clonidine	1mg/mL	50mg in 50mL	C / P
Dexmedetomidine	2mg/mL	100mg in 50mL	C / P
	15micrograms/mL	750micrograms in 50mL	C / P
Adrenaline	4micrograms/mL	200micrograms in 50mL	C / P
	80micrograms/mL	4mg in 50mL	C / P
	160micrograms/mL	8mg in 50mL	C / P
	320micrograms/mL	16mg in 50mL	C
	16micrograms/mL	4mg in 250mL	C / P
Noradrenaline	80micrograms/mL	4mg in 50mL	C
	160micrograms/mL	8mg in 50mL	C
	160micrograms/mL	16mg in 100mL	C
	320micrograms/mL	32mg in 100mL	C
Vasopressin (Argipressin)	0.4units/mL	20units in 50mL	C / P
Dobutamine	5mg/mL	250mg in 50mL	C
		500mg in 100mL	C

REFERENCES

([https:// doi. org/ 10. 1007/ s00134-021-06506-y](https://doi.org/10.1007/s00134-021-06506-y)) and Critical Care Medicine

([https:// doi. org/ 10. 1097/ CCM. 00000000000005337](https://doi.org/10.1097/CCM.00000000000005337))

Ouerd, S et al Vasopressin Use in the Support of Organ Donors: Physiological Rationale and Review of the Literature *Crit Care Explor.* 2023 Apr; 5(4): 0907

